

4.6 Triple Integrals in Cylindrical & Spherical Coordinates Worksheet

1. Let the point $P = (1, -\sqrt{3}, 2\sqrt{3})$ be given in rectangular coordinates. Express this point in terms of both cylindrical and spherical coordinates.

2. Set up, but do not evaluate, the integral $\iiint_E (x + y) dV$, where E is bounded by the xy -plane and the hemispheres $z = -\sqrt{4 - x^2 - y^2}$ and $z = -\sqrt{9 - x^2 - y^2}$.

3. Set up, but do not evaluate, a triple integration to find the volume of a solid E , where E is enclosed by $z = \sqrt{x^2 + y^2}$ and $z = 2$.

4. Consider the solid bounded by the cone $z = \sqrt{x^2 + y^2}$ and the sphere $x^2 + y^2 + z^2 = 1$.
- a) Use spherical coordinates to find the volume of this solid.

b) Use cylindrical coordinates to find the volume of this solid.

5. Consider the region E bounded by the xy -plane, the cylinder $x^2 + y^2 = 1$ and the plane $x + z = 2$.

Calculate the value of $\iiint_E (x^2 + 2 + y^2) dV$